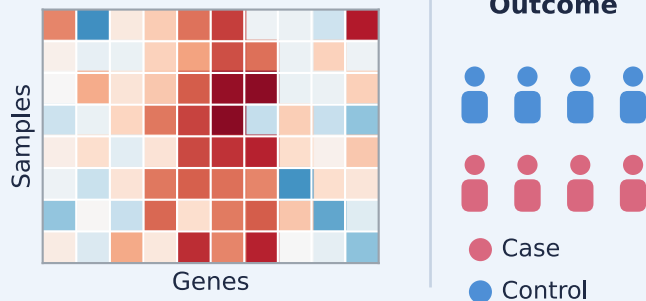
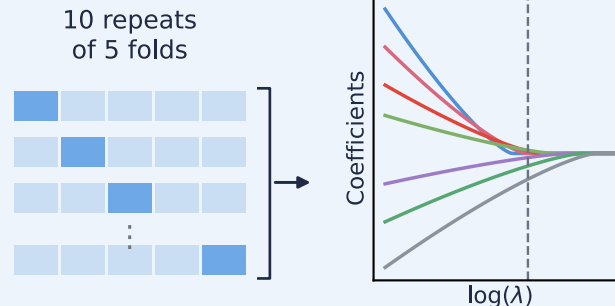


Training expression and outcome

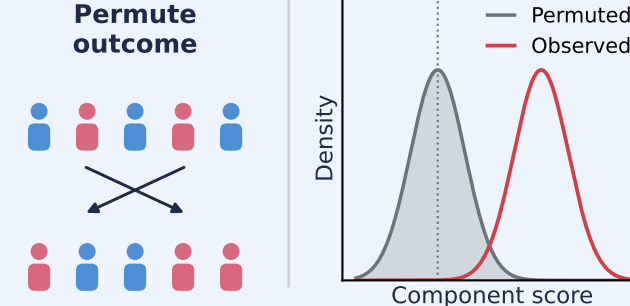


Train-only processing
2,000 variable genes

Repeated elastic-net models

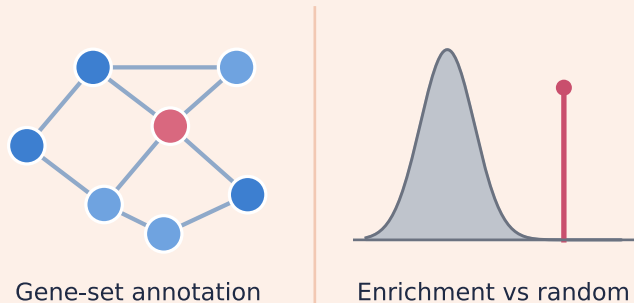


Outcome-permutation comparison



Biological annotation
does not alter ranking

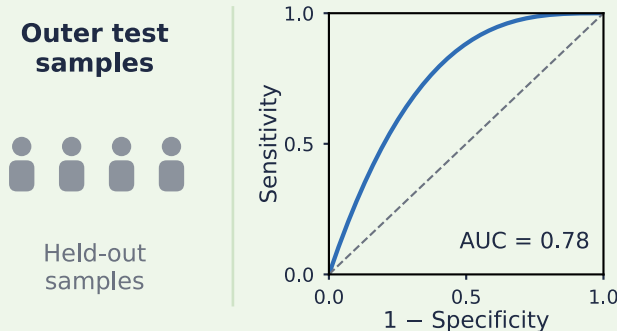
Interpret selected genes



GO, Hallmark, Open Targets
Compare with matched random sets

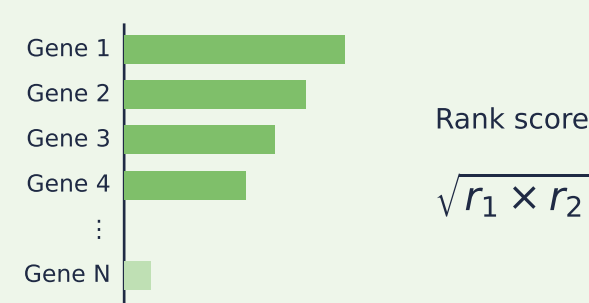
Outer test samples quantify
predictive performance

Evaluate the selected gene set



Common prediction procedure
Held-out outer samples

Rank genes



Geometric mean of the two ratios
Select a fixed or training-chosen size